

Hand Gesture Recognition

30 Days Complete Roadmap (Python + MediaPipe)

■ **Tools:** Python 3.7+ | OpenCV | MediaPipe | NumPy | Webcam **Install:** `pip install opencv-python mediapipe numpy`

■ Week 1: Environment Setup & MediaPipe Basics

Day 1

- Install Python, OpenCV, MediaPipe, NumPy
- Test webcam access with OpenCV
- Study `hand_gesture.py` file structure

Day 2

- Learn OpenCV basics: capture video, show frames
- Understand BGR color format
- Study `estimate_brightness()` and `apply_low_light_enhancement()`

Day 3

- Learn MediaPipe `HandLandmarker` basics
- Understand 21 hand landmarks
- Download `hand_landmarker.task` model file

Day 4

- Study landmark IDs: `THUMB_TIP=4`, `INDEX_TIP=8`, `MIDDLE_TIP=12`
- Understand `TIP_IDS`, `DIP_IDS`, `PIP_IDS`, `MCP_IDS` arrays
- Practice identifying each finger point

Day 5

- Learn `dist()` — 3D Euclidean distance
- Learn `dist_2d()` — 2D distance for U vs V detection
- Learn `angle_3pts()` and `palm_size` calculation

Day 6

- Study `is_finger_up()` — thumb vs other fingers
- Study `is_finger_curbed()` and `is_finger_hooked()`
- Practice detecting finger states with `get_finger_states()`

Day 7

- Review Week 1 concepts
- Test hand landmark detection stability
- Study `LetterSmoother` class — understand `buffer_size`

■ Week 2: Gesture Detection & Classification

Day 8

- Study `get_finger_states()` function
- Understand handedness (Left/Right) effect on thumb
- Practice detecting all 5 finger states

Day 9

- Study `thumb_beside_fist()` — A sign
- Study `thumb_is_across_palm()` — B sign
- Practice detecting A, B, S signs

Day 10

- Study `fingertips_touching_thumb()` — O sign
- Study `thumb_index_circle()` — F, D signs
- Practice detecting O, F, D, E signs

Day 11

- Study `hand_pointing_sideways()` — G, H signs
- Study `hand_pointing_down()` — P, Q signs
- Practice detecting G, H, P, Q signs

Day 12

- Study `index_middle_crossed()` — R sign
- Understand `idx_mid_tip` distance: U vs V
- Practice detecting R, U, V, K signs

Day 13

- Study `classify_asl_letter()` full function
- Understand the 11 classification groups
- Test each ASL letter A-Z one by one

Day 14

- Test all 26 ASL letters A-Z
- Note confused letters: M/N, U/V, A/S
- Document accuracy issues for next week

■ Week 3: UI & Visualization

Day 15

- Study `draw_hand_landmarks()` function
- Customize landmark colors and line thickness
- Understand responsive scale = $h / 480.0$

Day 16

- Study `draw_status_panel()` — top-left info panel
- Understand `draw_rounded_rect()` — transparent panels
- Practice modifying panel layout

Day 17

- Study `draw_reference_panel()` — right side guide
- Understand `ASL_DESCRIPTIONS` & `SIGN_INST`
- Add your name and custom text to the UI

Day 18

- Study `LetterSmoother` class in depth
- Test buffer sizes: 3, 5, 7, 10
- Understand 40% agreement threshold

Day 19

- Study the `main()` function complete flow
- Understand `HandLandmarkerOptions` settings
- Test different detection confidence values

Day 20

- Add word building with `collected_letters`
- Add space key to complete words
- Display current word on screen

Day 21

- Study `draw_settings_panel()` with `BG_PRESETS`
- Test full UI on different screen sizes
- Fix any UI scaling issues

■ Week 4: Improvement & Final Project

Day 22

- Improve accuracy for M, N, T signs
- Fine-tune: `close = palm_size * 0.3` threshold
- Test edge cases (hand too close or far)

Day 23

- Add word building feature
- Collect detected letters into words
- Add backspace gesture (hold fist 2 seconds)

Day 24

- Add quiz / practice mode
- Show random letter to practice
- Give feedback + hint from `SIGN_INSTRUCTIONS`

Day 25

- Add score tracking system
- Track correct/incorrect gestures
- Display accuracy percentage on screen

Day 26

- Performance optimization — add FPS counter
- Reduce detection lag
- Test on different lighting conditions

Day 27

- Error handling & bug fixes
- Handle webcam not found error
- Handle model file missing — show download link

Day 28

- Record demo video of the project
- Test all 26 letters on camera
- Document any remaining issues

Day 29

- Update `README.md` with new features
- Add screenshots to documentation
- Write project report

Day 30

- Final review & testing
- Push final version to GitHub
- Submit project! ■

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